

## ISPARTA SULEYMAN DEMIREL, Turkey

WMO: 172410

Lat: 37.855N

Lon: 30.368E

Elev: 864

StdP: 91.37

Time Zone: 3.00 (E03)

Period: 04-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -7.9       | -6.0 | -12.9                         | 1.4 | -4.5 | -11.0 | 1.6 | -3.1 | 12.2                  | 4.5  | 11.2 | 4.3  | 3.2                   | 340  | 0.535 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.5                   | 35.8            | 17.0 | 34.2 | 16.7 | 33.0 | 16.6 | 19.0                | 29.2 | 18.4 | 29.1 | 17.8 | 28.8 | 4.4                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.1                            | 12.7 | 21.9 | 15.0 | 11.9 | 21.8 | 14.1 | 11.1 | 21.2 | 57.8          | 29.6 | 55.5 | 29.0 | 53.6 | 28.9 | 22.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 10.5          | 9.2  | 8.1 | -11.2                      | 37.9 | 2.5                | 1.5 | -13.0                                              | 39.0 | -14.5      | 39.9 | -15.9      | 40.8 | -17.7      | 41.9 |    |    |
| (5)               |      |     | -11.8                      | 20.6 | 2.4                | 0.9 | -13.5                                              | 21.3 | -14.9      | 21.8 | -16.2      | 22.3 | -17.9      | 23.0 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 13.5   | 2.4   | 4.6   | 7.8   | 11.8  | 16.2  | 21.2  | 25.1  | 25.1  | 20.5  | 14.6  | 8.5   | 4.0   |  |  |
|                                                                        | DBStd       | 8.44   | 3.68  | 3.85  | 3.06  | 2.94  | 2.98  | 2.81  | 2.06  | 1.95  | 2.81  | 2.97  | 3.30  | 3.55  |  |  |
|                                                                        | HDD10.0     | 743    | 237   | 154   | 80    | 14    | 0     | 0     | 0     | 0     | 0     | 4     | 64    | 188   |  |  |
|                                                                        | HDD18.3     | 2360   | 496   | 386   | 327   | 196   | 79    | 5     | 0     | 0     | 12    | 119   | 295   | 446   |  |  |
|                                                                        | CDD10.0     | 2026   | 0     | 2     | 11    | 68    | 193   | 336   | 469   | 468   | 314   | 146   | 19    | 1     |  |  |
|                                                                        | CDD18.3     | 603    | 0     | 0     | 0     | 0     | 13    | 91    | 211   | 210   | 76    | 2     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 7904   | 0     | 0     | 0     | 43    | 299   | 1199  | 2655  | 2571  | 1032  | 104   | 1     | 0     |  |  |
|                                                                        | CDH26.7     | 3716   | 0     | 0     | 0     | 3     | 73    | 490   | 1389  | 1322  | 425   | 13    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 3.4    | 3.2   | 3.4   | 3.7   | 3.5   | 3.3   | 3.7   | 4.1   | 3.8   | 3.4   | 3.0   | 2.7   | 2.8   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 471    | 51    | 42    | 54    | 55    | 51    | 27    | 22    | 15    | 18    | 35    | 46    | 64    |  |  |
|                                                                        | PrecMax     | 620    | 166   | 81    | 137   | 134   | 117   | 75    | 59    | 59    | 83    | 126   | 122   | 209   |  |  |
|                                                                        | PrecMin     | 327    | 3     | 15    | 10    | 24    | 6     | 2     | 1     | 1     | 0     | 2     | 7     | 1     |  |  |
|                                                                        | PrecStd     | 78     | 34    | 19    | 33    | 28    | 26    | 17    | 19    | 14    | 20    | 30    | 28    | 49    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 13.1  | 17.2  | 20.8  | 26.9  | 30.8  | 35.2  | 38.1  | 37.1  | 34.7  | 28.2  | 21.7  | 15.8  |  |  |
|                                                                        |             | MCWB   | 8.0   | 9.3   | 10.4  | 13.7  | 15.4  | 17.0  | 17.7  | 17.7  | 15.9  | 13.8  | 12.0  | 9.7   |  |  |
|                                                                        | 2%          | DB     | 11.0  | 14.9  | 18.8  | 24.3  | 28.9  | 33.2  | 36.1  | 35.6  | 32.9  | 26.0  | 19.8  | 13.0  |  |  |
|                                                                        |             | MCWB   | 7.0   | 8.3   | 9.5   | 12.5  | 15.0  | 16.3  | 17.0  | 17.1  | 15.7  | 13.6  | 11.3  | 8.4   |  |  |
|                                                                        | 5%          | DB     | 9.8   | 12.8  | 16.9  | 22.1  | 26.9  | 31.8  | 34.2  | 34.1  | 31.1  | 24.1  | 17.2  | 11.2  |  |  |
|                                                                        |             | MCWB   | 6.4   | 7.5   | 8.7   | 11.9  | 14.5  | 16.0  | 16.9  | 16.9  | 15.1  | 13.1  | 10.3  | 7.5   |  |  |
|                                                                        | 10%         | DB     | 8.1   | 10.9  | 14.9  | 19.9  | 24.8  | 29.8  | 33.0  | 32.9  | 29.2  | 22.1  | 15.2  | 9.9   |  |  |
|                                                                        |             | MCWB   | 5.5   | 6.5   | 8.3   | 11.2  | 14.0  | 15.8  | 16.7  | 16.7  | 14.9  | 12.6  | 9.4   | 6.9   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 8.9   | 10.4  | 11.5  | 14.8  | 17.0  | 18.9  | 20.1  | 20.1  | 18.3  | 15.7  | 13.6  | 10.5  |  |  |
|                                                                        |             | MCDB   | 11.9  | 16.1  | 17.9  | 24.1  | 26.9  | 27.9  | 30.2  | 28.7  | 27.6  | 23.2  | 18.9  | 14.0  |  |  |
|                                                                        | 2%          | WB     | 7.5   | 8.8   | 10.4  | 13.2  | 16.0  | 17.9  | 18.9  | 19.2  | 17.3  | 14.8  | 12.2  | 9.0   |  |  |
|                                                                        |             | MCDB   | 10.6  | 13.4  | 16.5  | 21.8  | 25.5  | 28.1  | 29.9  | 29.1  | 26.9  | 21.5  | 17.6  | 12.4  |  |  |
|                                                                        | 5%          | WB     | 6.5   | 7.9   | 9.4   | 12.4  | 15.2  | 17.2  | 18.2  | 18.6  | 16.6  | 14.2  | 11.1  | 8.2   |  |  |
|                                                                        |             | MCDB   | 9.2   | 12.3  | 15.4  | 20.5  | 24.2  | 27.5  | 29.6  | 29.0  | 26.3  | 21.4  | 15.8  | 10.8  |  |  |
|                                                                        | 10%         | WB     | 5.6   | 7.0   | 8.6   | 11.4  | 14.4  | 16.6  | 17.7  | 18.0  | 15.9  | 13.5  | 10.2  | 7.2   |  |  |
|                                                                        |             | MCDB   | 7.7   | 10.5  | 14.1  | 18.7  | 22.8  | 26.7  | 29.4  | 28.8  | 25.8  | 20.5  | 14.7  | 9.5   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 9.0   | 10.5  | 12.0  | 13.8  | 14.3  | 15.2  | 15.5  | 15.6  | 15.8  | 14.3  | 13.0  | 9.7   |  |  |
|                                                                        |             | MCDBR  | 11.4  | 13.8  | 16.8  | 18.6  | 18.1  | 17.6  | 17.7  | 17.4  | 19.1  | 17.6  | 16.4  | 11.9  |  |  |
|                                                                        |             | MCWBR  | 7.6   | 8.2   | 9.1   | 8.8   | 7.2   | 5.7   | 5.6   | 5.6   | 7.1   | 7.9   | 9.2   | 7.8   |  |  |
|                                                                        |             | MCDBR  | 9.6   | 12.3  | 13.7  | 16.2  | 15.8  | 15.5  | 15.7  | 15.2  | 15.6  | 14.5  | 13.5  | 9.9   |  |  |
|                                                                        | 5% WB       | MCWBR  | 6.6   | 7.5   | 7.8   | 8.2   | 6.9   | 5.5   | 5.0   | 4.8   | 5.7   | 6.8   | 7.9   | 6.7   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.305 | 0.335 | 0.380 | 0.413 | 0.400 | 0.370 | 0.367 | 0.374 | 0.352 | 0.349 | 0.320 | 0.306 |  |  |
|                                                                        | taud        |        | 2.433 | 2.353 | 2.223 | 2.159 | 2.261 | 2.372 | 2.376 | 2.336 | 2.406 | 2.402 | 2.452 | 2.473 |  |  |
|                                                                        | Ebn at Noon |        | 878   | 890   | 876   | 866   | 882   | 906   | 906   | 893   | 897   | 863   | 853   | 852   |  |  |
|                                                                        | Edh at Noon |        | 84    | 104   | 130   | 147   | 136   | 121   | 120   | 122   | 107   | 98    | 82    | 76    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.17  | 3.01  | 4.28  | 5.42  | 6.38  | 7.39  | 7.66  | 6.79  | 5.54  | 3.95  | 2.78  | 2.00  |  |  |
|                                                                        | RadStd      |        | 0.28  | 0.37  | 0.32  | 0.44  | 0.46  | 0.47  | 0.24  | 0.20  | 0.33  | 0.24  | 0.24  | 0.29  |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
|      |                         | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (11 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page